Abhay Santhani

abhaysanthani@gmail.com | +91-8374162770 | linkedin.com/in/abhay-santhani | Hyderabad, India

PROFESSIONAL SUMMARY

Applied AI Engineer with 2+ years of experience building production LLM systems across pharma, finance, and enterprise automation. Specialises in end-to-end LLM data pipelines, agentic AI frameworks (LangGraph, AutoGen), and domain adaptation of large language models (3B–70B). Proven track record shipping live AI products — from a deployed finance intelligence platform used by real analysts to an agentic QA framework that eliminated the bulk of manual testing effort. Seeking to bring deep hands-on AI engineering experience to a high-impact product team.

EXPERIENCE

Tekframeworks Sep 2025 – Present

AI/ML Engineer — Product

FoRMLM | Pharma & Finance Domain LLM Data Infrastructure

- Architected an end-to-end LLM training data pipeline processing **200+ GB** of regulatory pharmaceutical corpora (**USFDA, EMA, ICH** guidelines) — transforming raw documents into LoRA fine-tuning-ready datasets for **3B–70B models**.
- Built a layout-aware PDF ingestion engine using **PyMuPDF (Fitz)** to parse complex regulatory PDFs and XML filings, preserving semantic structure (tables, sections, headers) critical for high-fidelity LLM training data.
- Designed a data quality framework covering **duplicate/near-duplicate detection** and **content sanity checks** (empty chunks, garbled text, encoding errors), ensuring contamination-controlled corpora at scale.
- Built an **S3-based data versioning** and lifecycle management pipeline to support reproducible, scalable dataset iterations across multiple domain adaptation experiments.
- Developed a **GPT-4o-powered synthetic SFT Q&A generation** pipeline to produce instruction-tuning datasets from regulatory source material, directly improving downstream DAPT model performance.
- Processed a sample PubMed corpus in parallel to validate pipeline generalisability across biomedical literature, establishing a cross-domain reuse pattern for future domains.

SparLM | Finance Analyst Intelligence Platform (Live Product)

- Co-built (2-engineer team) the full AI backend of SparLM — a **live platform** used by real financial analysts — enabling automated **Month-on-Month report generation** from pitch decks and earnings call transcripts.
- Designed the document retrieval and prompt-orchestration pipeline using **Qdrant vector database** with open-source **sentence-transformer embeddings**, enabling semantically accurate context retrieval across multi-document financial corpora.
- Built a cross-document inconsistency detection system powered by **Qwen 387B**, automatically surfacing contradictions between management guidance statements in call transcripts and filed annual reports.
- Delivered **structured PDF reports** as the system output — consumable directly by analysts and frontend, with source-traceable LLM-generated insights.
- Contributed to **PostgreSQL** schema design to support document metadata, analytical outputs, and end-to-end source-to-insight traceability.

Agivant Technologies Nov 2024 – Aug 2025

Machine Learning Engineer — Solutions

Executive Insight Generator | Agentic Finance AI (POC)

- Engineered the **LangGraph orchestration layer** of a multi-agent system integrating Gemini and GPT models to generate CXO-level financial insights from enterprise datasets, reducing manual analyst effort by **~90%**.
- Operationalised insight delivery via **Microsoft Fabric Data Agents**, enabling scalable real-time report generation — demonstrated to potential enterprise clients as a POC.

Agentic QA Automation Framework | E-commerce Web Testing

- Built an **end-to-end agentic QA framework** from scratch — agent logic, Selenium/Playwright browser automation, and test orchestration — using **LangGraph** to autonomously navigate and validate e-commerce web flows.
- Eliminated the majority of repetitive manual QA cycles, with the QA team estimating **~70% reduction** in manual testing effort after deployment.

Audio Intelligence & GenAI Query System | Sales Intelligence

- Built a context-aware audio intelligence system enabling natural-language search across hours of sales call recordings using automated transcription, semantic chunking, and GenAI querying — helping teams detect **buyer intent signals** at scale.

Geon
Apr 2024 – Oct 2024

Data Scientist Intern — R&D

- Owned the full scope of battery ML development: built SoC and State of Health (SoH) prediction models achieving **80%+ accuracy**, designed the ETL pipeline for battery sensor data, and created automated performance dashboards — all from scratch.
- Replaced a **month-long manual plant testing** process with an ML-powered inference pipeline that produces equivalent battery health assessments in **under 10 minutes** — a **~99% reduction** in cycle time that directly unblocked the R&D team.
- Built an ETL pipeline handling missing value imputation, outlier removal, and custom feature engineering across **20+ battery datasets**, directly improving model accuracy and stability.
- Delivered automated dashboards tracking real-time battery performance metrics, enabling data-driven decisions by plant management.

Indian Institute of Tropical Meteorology (IITM)
Jan 2024 – Mar 2024

Machine Learning Engineer Intern — R&D

- Deployed ML models with **82% accuracy** on a public meteorological platform, expanding access to weather research insights for **40% more users**.
- Implemented an ETL pipeline for raw meteorological data wrangling and feature engineering, improving model robustness on **High-Performance Computational Systems (HPCS)**.
- Restructured legacy ML models for forward compatibility, reducing model update cycles by **15%**.

PROJECTS

MemoryFolio
Python · PyTorch · Deep Learning

- Built a face recognition and clustering system using CNN architectures (**VGG, FaceNet**) in PyTorch, achieving **~85%** identification accuracy through iterative model tuning and data preprocessing on an academic dataset.

CERN Di-electron Collision Analysis
Python · Unsupervised ML

- Applied unsupervised learning techniques — Self-Organising Maps (SOM), Restricted Boltzmann Machines (RBM), and Isolation Forest — to the **CERN Open Data Portal**'s LHC collision dataset to detect anomalous patterns in high-energy particle events.

SKILLS

Languages: Python, SQL, Bash

LLM & Agentic Frameworks: LangGraph, AutoGen, Google ADK, MCP, Prompt Engineering, LoRA, DAPT, SFT

AI / ML: LLMs, Domain Adaptation, NLP, Deep Learning, Time Series, Supervised & Unsupervised Learning

Data Engineering: Medallion Architecture, ETL Pipelines, Vector Databases (Qdrant), Tokenisation, Sharding, S3

Frameworks & Tools: PyTorch, TensorFlow, PyMuPDF, Flask, Spark, Docker, Streamlit, Selenium, Playwright

Databases: PostgreSQL, MySQL, Qdrant

Cloud: AWS (S3), Azure (Microsoft Fabric), GCP

Visualisation: Tableau, Power BI

EDUCATION

Vijaybhoomi University — B.Tech in Artificial Intelligence
Sep 2020 – Oct 2024

CGPA: 8.76 / 10.0 | Awarded Dean's Medal for Outstanding Academic Performance

CERTIFICATIONS & ACHIEVEMENTS

- Qualified GATE Data Science & AI 2024
- **Salesforce AgentForce** — Corporate Certification (Salesforce)
- **Salesforce Cloud Data Admin** — Corporate Certification (Salesforce)
- Predictive Business Analytics — Coursera